



TELETRAFFIC

Introduction

- There are many telephone customers, much larger than the number of available trunks.
- Not every customer makes or receives a telephone call at the same time.
- The number of trunks connecting the Mobile Switching Center (MSC) *A* with another MSC *B* are the number of voice pairs used in the connection (duplex).

Cont...

- One of the most important steps in telecommunication engineering is to determine the number of trunks required on a route or a connection between MSCs.
- To dimension a route correctly, we must have some idea of its usage, that is, how many subscribers are expected to talk at one time over the route.
- The call capacity of an MSC is expressed as the maximum number of originating plus incoming (O I) calls that can be handled during a high-traffic hour while meeting the dial-tone delay requirements.

Traffic Usage

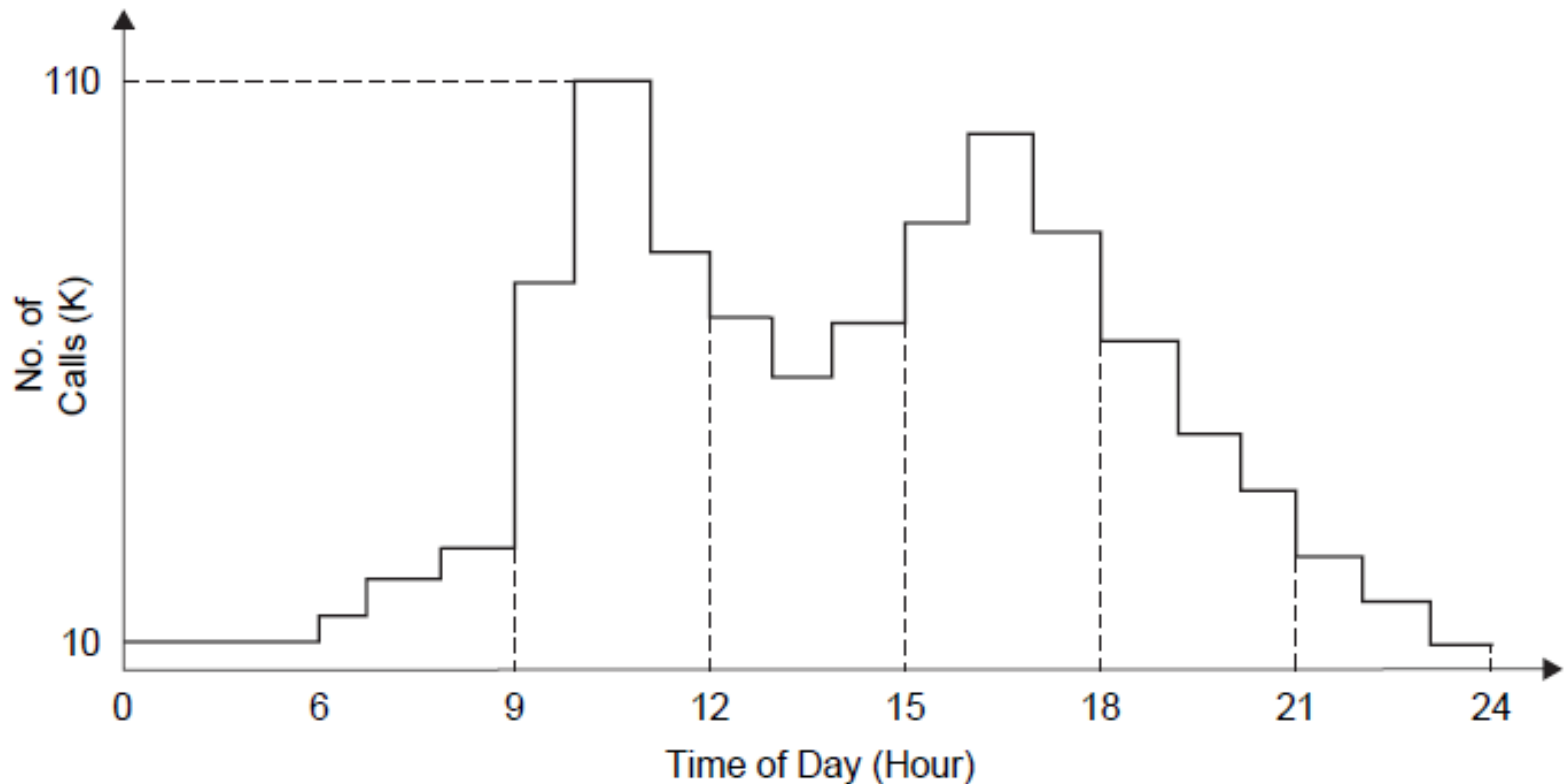
- A traffic path is a communication channel, time slot, frequency band, line, trunk, switch, or circuit over which individual communications take place in sequence.
- Traffic usage is defined by two parameters, *calling rate* and *call holding time*
- Calling rate, or the number of times a route or traffic path is used per unit time;
- Call holding time, or the average duration of occupancy of a traffic path by a call.

Cont...

- The *carried traffic* is the volume of traffic actually carried by a switch.
- *offered traffic* is the volume of traffic offered to a switch.
- *The offered load* is the sum of the carried load and overflow (traffic that cannot be handled by the switch).

$$\text{Offered load} = \text{carried load} + \text{overflow}$$

Example of voice traffic variation hour by hour.



Cont...

- The figure shows a typical hour-by-hour voice traffic variation for an MSC.
- We notice that the busiest period — the *busy hour (BH)* is between 10 A.M. and 11 A.M.
- Busy hour as the span of time (not necessarily a clock hour) that has the highest average traffic load for the business day throughout the busy season.
- The **peak hour** is defined as the clock hour with highest traffic load for a single day.

Cont...

- Since traffic also varies from month to month, we define the **average busy season (ABS)** as the three months (not necessarily consecutive) with the highest average BH traffic load per access line.
- Telephone systems are not engineered for maximum peak loads, but for some typical BH load.
- The blocking probability is defined as the average ratio of blocked calls to total calls and is referred to as the GoS.

Example

- If there are 380 seizures (lines connected for service) and 10 blocked calls (lost calls) during the BH, what is the GoS?

Solution

Blocking Probability = (Number of lost calls)/(Total number of offered calls)

$$\text{GoS} = 10 / (380 + 10) = 1 / 39$$

Traffic Measurement Units

- **Traffic intensity** is the average number of calls simultaneously in progress during a particular period of time.
- It is measured either in units of **Erlangs or CCS**.
- An average of one call in progress during an hour represents a traffic intensity of 1 Erlang.
- thus 1 Erlang equals 1×3600 call seconds (36 CCS).
- The Erlang is a dimensionless number.

Cont...

- Traffic intensity can be obtained as:

$$\text{Traffic Intensity} = \frac{\text{(the sum of circuit holding time)}}{\text{(the duration of monitoring period)}}$$

$$I = \frac{\sum_{i=1}^{N_c} t_i}{T} = \frac{N_c \bar{t}}{T} = n_c \bar{t}$$

where:

I = traffic intensity

T = duration of monitoring period

t_i = the holding time of the i th individual call

N_c = the total number of calls in monitoring period

$\bar{t}$ = average holding time

n_c = number of calls per unit time

Cont...

- Percentage of occupancy is the Percentage of time a server is busy.
- Peg count is the number of attempts to use a piece of equipment.
- The usage (U), *peg count (PC) per time period, overflow (O) per period, and average holding time t are related as:*

$$U = (PC - O) \cdot \bar{t}$$

Example

- In a switching office an equipment component with an average holding time of 5 seconds has a peg count of 450 for a one-hour period. Assuming that there was no overflow (i.e., the system handled all calls), how much usage in call-seconds, CCS, and Erlangs has accumulated on the piece of the equipment?

Soln

$$U = (450 - 0) \times \frac{5}{3600} = 0.625 \text{ Erlangs}$$

$$0.625 \text{ Erlangs} \times 36 \text{ CCS/Erlangs} = 22.5 \text{ CCS} = 2250 \text{ call-seconds}$$

Example

- In a wireless network each subscriber generates two calls per hour on the average and a typical call holding time is 120 seconds. What is the traffic intensity?
- Solution

$$I = \frac{2 \times 120}{3600} = 0.0667 \text{ Erlangs} = 2.4 \text{ CCS}$$

Example

- In order to determine voice traffic on a line, we collected the following data during a period of 90 minutes as shown in the table. Calculate the traffic intensity in Erlangs and CCS.

Call no.	Duration of call (s)
1	60
2	74
3	80
4	90
5	92
6	70
7	96
8	48
9	64
10	126

Soln

$$\text{Call arrival rate} = \frac{10}{1.5} = 6.667 \text{ calls/hour}$$

Average call holding time:

$$\bar{t} = \frac{(60 + 74 + 80 + 90 + 92 + 70 + 96 + 48 + 64 + 126)}{10} = 80 \text{ seconds per call}$$

$$I = \frac{6.667 \times 80}{3600} = 0.148 \text{ Erlangs} = 5.33 \text{ CCS}$$

Grade of Service (GoS)

- Is the measure of probability that a percentage of the offered traffic will be blocked or delayed.
- Commonly expressed as the fraction of calls that fails to receive immediate services.
- For example if GoS is 0.05, then one call out of 20 will be blocked during busy hour.

Grade of Service (GoS)

- ▶ Grade of Service, B

$$B = \frac{\text{Lost traffic}}{\text{Offered traffic}}$$

Or

$$B = \frac{\text{Lost call}}{\text{Offered call}}$$

Example

1200 calls are offered to a channel and 6 calls are lost. Duration of a call is 3 minutes. Find:

- a) Offered traffic, A
- b) Carried traffic
- c) Lost traffic
- d) GOS, B
- e) Congestion time

Solution:

a) $A = CH/T = (1200 \times 3) / 60 = 60 \text{ E}$

b) Carried traffic = $[(1200-6) \times 3] / 60 = 59.7 \text{ E}$

c) Lost traffic = $(6 \times 3) / 60 = 0.3 \text{ E}$

d) $B = \text{Lost traffic} / \text{Offered traffic}$
 $= 6/1200$
 $= 0.005$

e) Congestion time = $B \times 1 \text{ hours (second)}$
 $= 0.005 \times 60 \times 60$
 $= 18 \text{ seconds}$



END